RUTURAJ DESAI

Data Scientist | Machine Learning & Applied Analytics

+91-9879148289 | desairuturaj98@gmail.com | Ahmedabad, Gujarat, India | GitHub | LinkedIn

Summary

Data science and applied-ML professional with nearly 3 years of experience in AML analytics and reporting, complemented by end-to-end machine learning projects. Experienced in EDA, feature engineering, clustering, regression, classification, model evaluation, and visualization using Python, SQL, scikit-learn, XGBoost, LightGBM, Matplotlib, Seaborn, and Plotly. Built reproducible ML pipelines and FastAPI applications packaged with Docker, using MLflow, DVC, and AWS S3; translates model results into measurable operational improvements.

Technical Skills

- **Programming & Data:** Python (Pandas, NumPy), SQL (CTEs, joins, window functions), data cleaning and preprocessing
- **Classical ML:** Regression, classification, clustering, K-means, Random Forest, XGBoost, LightGBM, feature engineering, hyperparameter tuning
- **Statistics & Evaluation:** EDA, descriptive/statistical analysis, hypothesis testing, cross-validation, MAE, R^2, PR-AUC, lift analysis, silhouette score
- **Visualization:** Matplotlib, Seaborn, Plotly, Tableau, Power BI, Streamlit
- **ML Engineering & Cloud:** scikit-learn, FastAPI, Flask, Docker, MLflow, DVC, AWS S3, SHAP, LIME, Optuna
- **Domain:** AML transaction monitoring, sanctions screening, KYC/entity resolution, FINTRAC regulatory reporting

Professional Experience

Data Scientist (Functional) | AML Reporting Analyst **Feb 2024 – Sep 2026**

BMO Financial Group, Toronto

- Applied EDA and feature engineering to 45,000 historical cases and built a K-means clustering model (k-means++ initialization; silhouette-based selection) to segment case complexity using case-folder metadata.
- Developed and evaluated XGBoost and LightGBM regression models to estimate case-handling time, using complexity tiers as features and achieving an 8-minute MAE on held-out data.
- Collaborated on exposing model predictions through FastAPI applications packaged with Docker; DevOps colleagues supported scaling. The workflow contributed to 25% higher daily investigator throughput (8 to 10 cases) and a 2% relative reduction in SLA breaches.
- Contributed logic and testing to an RPA-supported sanctions reporting workflow, reducing per-case processing effort from approximately 3–4 hours to about 30 minutes, with investigator review retained.
- Produced 40+ FINTRAC reports weekly with under 3% QC variance and delivered 20+ executive dashboards using Tableau and Power BI to communicate operational and compliance insights.

Selected Machine Learning Projects

AML Transaction Detection — Production ML Pipeline **GitHub**

- Processed 5.08M synthetic transactions and engineered 59 base and graph/network features using strictly prior 1-, 3-, and 7-day windows to reduce temporal leakage.
- Tuned an XGBoost classifier using time-aware validation; achieved 0.572 validation PR-AUC and 0.396 held-out test PR-AUC, delivering 31.5× test lift over a seven-rule baseline.
- Built a seven-stage DVC pipeline and served inference with FastAPI and Pydantic; used MLflow for experiment tracking and model registry management.

Flight Price Prediction — End-to-End MLOps **GitHub**

- Built a reproducible DVC pipeline using AWS S3 remote storage; tuned Random Forest and XGBoost with Optuna, achieving $R^2 \approx 0.92$ and MAE ≈ 652.
- Deployed a Flask API that reapplies feature transformations at inference to maintain training-serving consistency.

Education

- **Postgraduate Diploma (PGD) — Business Analytics**, Conestoga College, Canada 2023
- **Postgraduate Diploma (PGD) — Project Management**, Conestoga College, Canada 2022
- **B.Tech, Engineering**, Indian Institute of Technology Ram (IITRAM), Ahmedabad, India 2020